

Module 14 Capstone Paper Outline + Progress Update

Pruned study plan: evidence-gated LLM alerts for pharmacogenomic and genomic claims

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Submission thesis: Stage 1 is pruning. The prior prototype showed implementation consistency across a broad synthetic assurance suite; the capstone-facing paper now narrows to one clinically interpretable workflow, three evidence gates, and an independent-review evaluation plan.

Pruned Working Title

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims: Detecting Overclaiming Relative to Guideline-Supported Evidence

The project asks whether a pre-display evidence gate can make LLM-generated pharmacogenomic or genomic medication-alert text less likely to overstate what the cited evidence supports. The claim is deliberately modest: Stage 1 evaluates specification conformance on author-designed synthetic archetypes, not clinical outcome improvement.

Research Question and Hypothesis

Research question: When a medical LLM drafts a pharmacogenomic or genomic medication alert, can an evidence-gated controller reduce overclaiming relative to guideline-supported evidence while tracking inappropriate denial?

Hypothesis: Compared with ungated LLM output, gated output will produce fewer statements that inflate association, surrogate endpoints, population transport, or citation strength into clinical action. The main planned outcomes are overclaim rate, inappropriate denial rate, calibration of claim strength to evidence level, and qualitative error categories.

Pruning Map

Decision	Kept for the paper	Deferred or boundary case
Workflow	Pharmacogenomic/genomic medication-alert text, especially where CPIC/PharmGKB-style evidence grades can anchor claims.	Other clinical workflows are outside this submission-facing paper.
Gates	Endpoint/actionability, population fit, and citation/guideline support.	Other mathematical, privacy, and governance analogies are outside this submission-facing paper.
High-risk examples	Opioid pharmacogenomics stays only where metabolism/safety guidance is guideline-grounded, such as CYP2D6-mediated codeine/tramadol language.	Claims that genetics can predict addiction liability for opioids or stimulants are treated as red-zone overclaim tests, not as clinical recommendations.
Boundary interventions	Included only as unsupported-action stressors when an association is inflated into an intervention claim.	No wet-lab, gene-editing, or treatment intervention is proposed.

Progress to Date

- Built a reproducible synthetic evidence-gating harness with stress-test cases and deterministic action selection.
- Current local verification: 7 focused tests pass for the standalone pruned evidence-gate evaluator, including a regression test that flips an expected-action label to confirm the controller output is label-independent.
- Current specification-conformance summary: 30/30 action conformance and 28/30 primary-gate conformance on author-designed scenarios. PGX19 and PGX24 reveal citation-first gate precedence.
- Current scope audit: the reviewer-facing package keeps one workflow, three gates, one synthetic case bank, and a Stage 2 independent-review plan.
- Important limitation: these results demonstrate implementation consistency against author-designed synthetic archetypes. The controller consumes structured evidence annotations and does not yet extract them from LLM text, citations, or EHR context.

Paper Outline

1. Abstract: concise statement of the overclaiming problem, the pruned workflow, the three gates, and the Stage 1 specification-conformance scaffold.
2. Introduction: explain why fluent medical LLM outputs can sound stronger than the evidence, connecting the course themes of surrogate endpoints, hard outcomes, clinician authority, and adoption.

3. Background: summarize pharmacogenomic alerting, guideline/evidence grading, citation hallucination, population transport, and the difference between association, mechanism, and intervention.
4. Methods: create independently authored alert cases; generate ungated LLM drafts; pass drafts through endpoint/actionability, population-fit, and citation/guideline-support gates; have blinded reviewers classify overclaims and inappropriate denials.
5. Results: report overclaim reduction, inappropriate denial, calibration, interrater agreement, and error categories for Stage 2. Present the existing synthetic result only as Stage 1 specification conformance.
6. Discussion: identify where gated LLMs can improve claim discipline, where clinical outcome trials would still be required, and why pruning is a safety feature rather than a retreat.

Immediate Empirical Plan

The final paper will replace the broad 30-scenario demonstration with a smaller independent evaluation. The planned case set will contain guideline-supported pharmacogenomic alerts, moderate or population-limited claims, unverifiable citation claims, and red-zone genomic claims such as addiction-liability or CRISPR actionability statements.

Each case will be evaluated in two arms: ungated LLM output and gated output. Reviewers will be blinded to arm when feasible and will mark whether the alert overclaims evidence, denies an appropriate claim, preserves clinician authority, and states the correct population/endpoint limits. If clinical reviewers are unavailable for pharmacogenomics, the fallback evaluation will be strictly guideline-text adjudication rather than clinical recommendation adjudication.

Why This Version Is Cohesive

The project no longer tries to prove a universal theory of medical AI assurance. It studies one recurring failure mode: LLMs compress weak, noisy, population-limited, or surrogate evidence into authoritative clinical language. Biophysics and systems biology remain useful background because they sharpen the distinction between measurement, mechanism, and intervention. They do not add a new branch to the methods.

Pruning is therefore the main methodological move: connections that are interesting but not necessary for the defined workflow are left out of the primary paper. They can appear, at most, as future work or as brief examples explaining why the gate must remain conservative.

Generative AI Use

Generative AI tools were used to brainstorm candidate framings, critique overbreadth, draft text, and help organize the artifact. The student selected the final scope, checked sources, ran local verification, and remains responsible for all submitted claims.

Key Sources Checked

Checked anchors as of August 2, 2026: CPIC/ClinPGx pharmacogenomic guideline materials; PharmGKB/ClinPGx evidence-level documentation; CDC NCHS overdose mortality data; CDC MMWR stimulant-involved overdose analysis; Bialek's biophysics text; and Alon's systems-biology design-principles text.